

ENV S 193DS Final

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[Github Repo] (https://github.com/nvanderpoel-cmd/ENVS-193DS_spring-2026_final)

Set up

```
# reading in packages  
library(here)
```

here() starts at /Users/nikivanderpoel/ENVS-193DS/GIT/ENVS-193DS_spring-2026_final/ENVS-193DS_spring-2026_final

```
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --  
v dplyr      1.2.1      v readr      2.2.0  
v forcats    1.0.1      v stringr    1.6.0  
v ggplot2    4.0.2      v tibble     3.3.1  
v lubridate  1.9.5      v tidyr      1.3.2  
v purrr      1.2.1
```

```
-- Conflicts ----- tidyverse_conflicts() --
```

```
x dplyr::filter() masks stats::filter()
```

```
x dplyr::lag()     masks stats::lag()
```

```
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```
library(janitor)
```

Attaching package: 'janitor'

The following objects are masked from 'package:stats':

chisq.test, fisher.test

```
library(dplyr)

# reading in data
nests <- read.csv(here("data/nest_data_final.csv"))
```

Problem 1. Research Writing

- a. Transparent statistical methods
- b. Suggestions for rewriting
- c. Further analysis

Problem 2. Data analysis

- a. Clean and wrangle

```
# new object nests_clean
nests_clean <- nests |>
  select(height_cm, case_control, ant_species) |>
  filter(ant_species %in% c(
    "AB",
    "RRB",
    "BBR")) |>
  mutate(ant_species = as_factor(ant_species),
         ant_species = fct_relevel(ant_species,
                                   "AB",
                                   "RRB",
                                   "BBR")) |>
  mutate(species_name = case_match(
    ant_species,
    "AB" ~ "Crematogaster sjostedti",
```

```
"RRB" ~ "Crematogaster mimosae",
"BBR" ~ "Crematogaster nigriceps"
))
```

Warning: There was 1 warning in `mutate()`.

i In argument: `species\_name = case\_match(...)`.

Caused by warning:

! `case\_match()` was deprecated in dplyr 1.2.0.

i Please use `recode\_values()` instead.

```
slice_sample(nests_clean, n = 10)
```

	height_cm	case_control	ant_species	species_name
1	163.5	0	RRB	Crematogaster mimosae
2	176.0	0	RRB	Crematogaster mimosae
3	154.0	0	RRB	Crematogaster mimosae
4	169.0	0	RRB	Crematogaster mimosae
5	423.0	0	RRB	Crematogaster mimosae
6	164.0	0	RRB	Crematogaster mimosae
7	216.0	0	RRB	Crematogaster mimosae
8	131.0	0	BBR	Crematogaster nigriceps
9	167.0	0	BBR	Crematogaster nigriceps
10	142.0	0	RRB	Crematogaster mimosae

- b. Response variable**
- c. Purpose of study**
- d. Table of models**
- e. Run the models**
- f. Select the best model**
- g. Check the diagnostics**
- h. Visualize the model predictions**
- i. Write a caption for your figure**
- j. Calculate model predictions**
- k. Interpret your results**

Problem 3. Affective and exploratory visualizations

- a. Comparing visualizations**
- b. Designing an analysis**
- c. Check any assumptions and run your analysis**
- d. Create a visualization**
- e. Write a caption**
- f. Write about your results**
- g. Sharing your affective visualization**